

# THANEESH BEDADHAM

1702 S Terracina Drive, Dublin, California 94568

☎ 925-548-3537 ✉ [tbedadham@ucsd.edu](mailto:tbedadham@ucsd.edu) 🔗 [linkedin.com/in/thaneesh-bed](https://www.linkedin.com/in/thaneesh-bed) 🐙 [github.com/tbedadham](https://github.com/tbedadham)

## Education

University of California, San Diego

B.S. in Mathematics - Computer Science

Expected June 2027

La Jolla, CA

## Relevant Coursework

- Data Structures
- Algorithms
- Discrete Mathematics
- Linear Algebra
- Computer Systems
- Probability & Statistics
- Multivariable Calculus
- Computational Methods

## Experience

### Software Engineer Intern

July 2025 – September 2025

Upx Labs

Remote

- Developed an AI-driven **conversation-to-documentation** pipeline that converted doctor-patient voice conversations into structured documented summaries, automating up to **80%** of manual note-taking tasks.
- Built an end-to-end Python pipeline using Whisper for transcription and LLM-based parsing, achieving **92% transcription accuracy** and consistent JSON structuring across 50+ sessions.
- Optimized schema validation and API modularity to reduce parsing errors by **35%** and enable scalability for healthcare and accessibility applications.

### AI Software Engineer Intern

September 2025 – Now

Outamation (via Extern)

Remote

- Built and evaluated early-stage **AI document-processing pipelines** using OCR, PyMuPDF, and pdfplumber to extract structured data from PDFs, reducing preprocessing time by **50%** across 1,000+ files.
- Developed a Retrieval-Augmented Generation (RAG) **chatbot** using Gradio and Python, enabling users to query PDF content through a web interface and receive LLM-based, context-grounded answers in real time.
- Implemented embedding-based retrieval workflows (LlamaIndex) to **improve response accuracy by 42%** and modularized pipeline components for scalable multi-document integration.

## Projects

### Minecraft Biome Classifier | Python, OpenCV, XGBoost, Computer Vision

2025

- Built and trained a supervised machine learning model from **1,500+ images** to classify Minecraft biomes from real gameplay images using domain-engineered color and texture features.
- Led model benchmarking and hyperparameter tuning efforts in a two-person team, comparing models and selecting XGBoost to improve baseline accuracy from the low 70% range to **nearly 90%**.

### Command-Line Text Editor | C, File I/O, Data Structures

2023

- Developed a **lightweight text editor in C** supporting file creation, editing, and navigation with manual memory management and real-time buffer updates.
- Optimized cursor traversal and allocation logic, reducing edit latency by **35%** for 10,000+ line files and achieving **zero memory leaks** verified via Valgrind.

### Neural Network from Scratch | Python, NumPy, Linear Algebra, Optimization

2024

- Derived and implemented a **neural network** in vanilla Python with forward/backpropagation for multi-layer perceptrons from first principles using matrix calculus, validating gradients via numerical checking.
- Implemented SGD and momentum-based optimization, achieving **90%+ accuracy** on benchmark classification tasks without ML frameworks.

### Tokenizer and Dictionary System | C, Hash Tables, AVL/BST Trees

2024

- Built a **custom tokenizer and dictionary manager** with  $O(1)$  lookup and  $O(\log n)$  ordering for 10,000+ tokens using hash tables and AVL trees.
- Ensured **100% memory efficiency** and zero segmentation faults under stress tests with Valgrind debugging and dynamic memory safety checks.

## Technical Skills

**Languages:** Python, C, C++, HTML, SQL, Assembly, MATLAB

**Tools:** OpenCV, XGBoost, PyTorch, Git, VS Code, Valgrind, Makefile, Linux/Unix, Jupyter Notebook

**Technologies:** FastAPI, REST APIs, MySQL, Whisper, Ollama, LlamaIndex, Gradio, PyMuPDF, Swift, pandas, NumPy

**Concepts:** Data Structures & Algorithms, Object-Oriented Programming, ML Fundamentals, Computer Vision,

Databases, Supervised Learning, Feature Engineering